

Gastric Tumours

Discipline: Pathology · Block M3 (Digestion & Defence) | Source: 8-
_Gastric__tumors_202627_DD3_1.pdf (29 slides) · Refs: Robbins, Cotran & Kumar 11th ed.; AMBOSS

Legend: ★ = high-yield · [added] = not on slides. Text **highlighted on the slides** (yellow/orange/underlined) is treated as lecturer emphasis and starred. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

Gastric tumours at a glance

Adenoca: **INTESTINAL**

Origin: precursors (dysplasia, adenoma), H. pylori
Site: antrum, lesser curve
Gross: bulky, exophytic or ulcerated (heaped borders)
Micro: GLANDS, apical mucin, mucin lakes, desmoplasia
Genes: CIN (APC, SMAD4/2, TP53), MSI (MLH1), EBV
Who: high-risk areas, mean 55 y, M:F 2:1

Adenoca: **DIFFUSE**

Origin: NO known precursor
Incidence uniform worldwide
Gross: no mass; stiff, thick wall, flat rugae → **linitis plastica**
Micro: SIGNET-RING cells, discohesive, NO glands
Gene: CDH1 (E-cadherin) loss
hereditary: α -catenin
Genomically stable, fewer mutations

GIST

Most common mesenchymal tumour of abdomen
50% in stomach · peak 60 y
From interstitial cells of **Cajal (pacemaker)**
KIT (75%) / PDGFRA (8%)
→ responds to imatinib
Gross: solitary, whorled, up to 30 cm
Micro: spindle / epithelioid
IHC: CD117 (KIT)
↑ with NF1

NET (carcinoid)

Well-differentiated NET; high grade = NE carcinoma
>40% in small intestine
Stomach: with atrophic gastritis, MEN1, ZES
Gross: yellow-tan, firm submucosal nodule
"Salt & pepper" chromatin
Synaptophysin + chromogranin
Foregut rarely spreads · midgut aggressive · hindgut almost benign

Adenocarcinoma metastases (often present at diagnosis)

Virchow node (left supraclavicular) · Sister Mary Joseph nodule (umbilicus) · Krukenberg (ovaries)
left axillary node (Irish node [name added]) · pouch of Douglas (Blumer shelf [name added])

Added diagram: the four gastric tumours compared

1. Title / Objectives

Objective	Likely tested as
★ Describe aetiology, pathogenesis, morphology and clinical features of gastric neoplasms	Intestinal vs diffuse adenocarcinoma; H. pylori; CDH1; linitis plastica; signet-ring; metastatic eponyms; GIST (KIT/CD117, Cajal); NET (salt-and-pepper, synaptophysin/chromogranin, carcinoid syndrome)

Flow: Introduction → Adenocarcinoma (epidemiology → aetiology → pathogenesis → morphology → clinical features → metastases) → GIST → Neuroendocrine tumours (carcinoid) → Summary → Case.

2. Main Content (Lecture Order)

2.1 Introduction (Slide 3)

- Gastric cancer = malignant transformation of gastric cells; **hereditary or sporadic**
- ★ **Types: adenocarcinoma, MALT lymphoma, leiomyosarcoma, GIST, NET**

- ★ **Adenocarcinoma = most common gastric malignancy (> 90%)**
- ★ Two morphological types:
 1. **Intestinal**: forms **bulky masses**
 2. **Diffuse**: **infiltrates and thickens** the wall
- ★ **Early symptoms** mimic chronic gastritis: **dyspepsia, dysphagia, nausea**
- ★ Usually found **late**, with **weight loss, anorexia, altered bowel habits, anaemia, haemorrhage**

🔄 **Recap: Intro.** Adenocarcinoma is > 90% of gastric cancers, split into intestinal (mass) and diffuse (infiltrative). Early symptoms look like gastritis, so diagnosis is usually late.

🔗 **Link:** First, who gets it and why.

▶ **Up next: Epidemiology and aetiology.**

2.2 Adenocarcinoma: Epidemiology & Aetiology (Slides 4-6)

Epidemiology:

- ★ Incidence varies **markedly by geography**: ★ **Japan**, Chile, Costa Rica, Eastern Europe up to ★ **20× higher** than North America, northern Europe, Africa, SE Asia
- Peak **55-80 y**; rare < 50
- ★ **M:F = 2-3 : 1**

Aetiology:

- ★ Most sporadic cancers relate to ★ **H. pylori**
- ★ **Chronic H. pylori = the main cause of non-cardia (distal) gastric cancer**
- ★ **EBV** (~10% of gastric cancers)
- ★ **Smoking**
- ★ **Heavy alcohol**

Other proposed risk factors: diet (**salt, preserved meats**, coffee) · low socioeconomic status · obesity · **pernicious anaemia** · **previous gastric surgery** · race/ethnicity · **blood group A**

🔄 **Recap: Who and why.** Japan-type high-risk regions, men (2-3:1), 55-80 y. H. pylori is the main cause of distal cancer; also EBV, smoking, alcohol, salty/preserved food, pernicious anaemia, prior gastric surgery, blood group A.

🔗 **Link:** The two histological types arise through different molecular routes.

▶ **Up next: Pathogenesis.**

2.3 Adenocarcinoma: Pathogenesis (Slides 7-8)

★★ Diffuse type:

- ★ **Loss-of-function mutation of the tumour-suppressor gene CDH1: ~30% of hereditary and ~50% of sporadic** diffuse gastric cancer
- ★ **CDH1 encodes E-cadherin** (cell-adhesion protein); ★ **E-cadherin loss is the key step** → cells become **discohesive** [link to morphology added]
- Hereditary diffuse cancer: **α-catenin mutation**
- ★ Diffuse tumours are **genomically stable** with **fewer mutations** than intestinal type

★★ Intestinal type: 3 molecular subtypes:

Subtype	Genes / features	Site
★ Chromosomal instability (CIN)	APC, SMAD4, SMAD2, TP53	★ Antrum and pyloric channel
★ Microsatellite instability (MSI)	★ MLH1 silencing	★ Gastric body
★ EBV-associated	Silencing of CDKN2A, PIK3CA ; ★ ↑ PD-L1/2	★ Fundus and proximal body

[added: ↑ PD-L1/2 makes EBV-associated tumours candidates for immune-checkpoint inhibitors.]

🔗 **Recap: Pathogenesis.** Diffuse = CDH1/E-cadherin loss (α -catenin in hereditary), genomically stable. Intestinal = CIN (APC, SMAD, TP53; antrum), MSI (MLH1; body), EBV (CDKN2A, PIK3CA, PD-L1; fundus).

🔗 **Link:** Molecules explain the appearance: cohesive glands vs single loose cells.

🔗 **Up next: Morphology.**

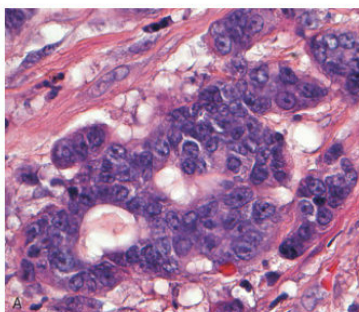
2.4 Adenocarcinoma: Morphology (Slides 9-12)

★ Intestinal type:

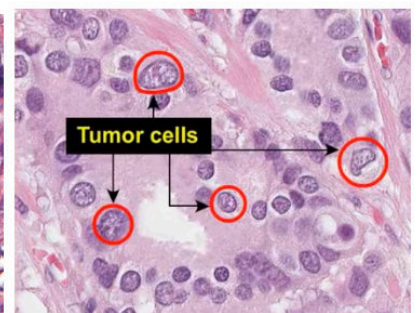
- Most distal cancers arise in the ★ **antrum**; ★ **lesser curvature > greater curvature**
- ★ **Bulky, gland-forming**, grow along **broad cohesive fronts**
- **Exophytic mass** or **ulcerated infiltrative tumour** (heaped-up borders, central ulcer)
- Cells have **apical mucin vacuoles**; mucin in gland lumina; **extracellular mucin lakes**
- Micro: **columnar, gland-forming cells infiltrating desmoplastic stroma**



Gastric adenocarcinoma. (A) Intestinal-type adenocarcinoma consisting of an elevated mass with heaped-up borders and central ulceration.



columnar, gland-forming cells infiltrating through desmoplastic stroma



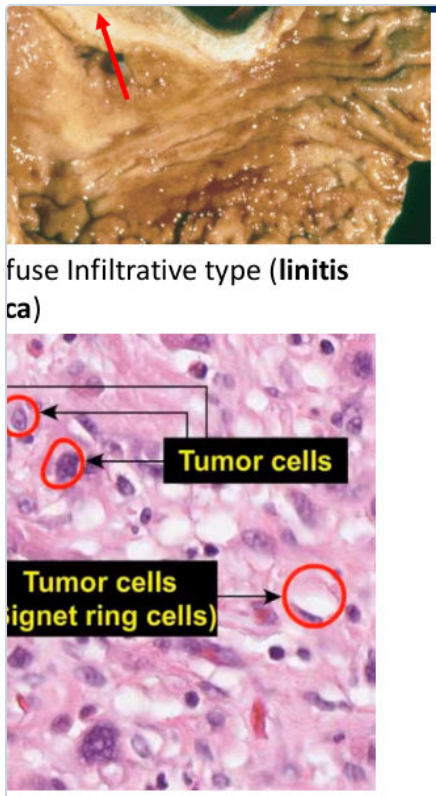
Tumor cells

Slide 10: Intestinal type: ulcerated mass with heaped-up borders; gland-forming tumour cells in desmoplastic stroma

★ Diffuse type:

- **Hard to see a mass**
- Infiltration provokes a **desmoplastic reaction** → **stiff wall**
- Large areas → ★ **flat rugae + rigid thick wall** = "**leather bottle stomach**" (linitis plastica)

- Micro: ★ **signet-ring cells**: large **mucin vacuole pushing the nucleus to the periphery**; **small clusters / single discohesive cells**; ★ **no glands**; mucin lakes



Slide 11: Linitis plastica (thick rigid wall) with infiltrating tumour cells



Slide 12: Diffuse type: thickened wall and signet-ring cells

🔄 **Recap: Morphology.** Intestinal = antrum/lesser curve, bulky or ulcerated, glands. Diffuse = no mass, desmoplasia, linitis plastica, signet-ring cells, no glands.

🔗 **Link:** Morphology also predicts who gets it and how it behaves.

👉 **Up next: Clinical features and prognosis.**

2.5 Adenocarcinoma: Clinical Features, Prognosis, Metastases (Slides 13-15)

	Intestinal	Diffuse
Geography	★ Predominates in high-risk areas	★ Uniform across countries
Precursors	★ Flat dysplasia, adenomas	★ None identified
Age / sex	★ Mean 55 y, M:F 2:1	

★★ Prognosis (Slide 14):

- ★ **Depth of invasion + extent of nodal and distant metastases = the most powerful prognostic indicators**
- Local invasion into **duodenum, pancreas, retroperitoneum** is common
- ★ **Early gastric cancer + resection → 5-year survival > 90%, even with nodal metastases**
- ★ **Advanced gastric cancer → 5-year survival < 20%**

★★★★ Metastases (often present at diagnosis) (Slide 15):

Site	Eponym
★ Supraclavicular sentinel node	★ Virchow node <i>[added: left side]</i>
★ Periumbilical node	★ Sister Mary Joseph nodule
★ Ovaries	★ Krukenberg tumour
Left axillary node	<i>[added: Irish node]</i>
Pouch of Douglas	<i>[added: Blumer shelf, felt on rectal exam]</i>

🔄 **Recap: Clinical.** Intestinal = high-risk areas, precursors, 55 y, men. Diffuse = everywhere, no precursor. Stage (depth, nodes, metastases) rules prognosis: early > 90% vs advanced < 20% at 5 years. Know Virchow, Sister Mary Joseph, Krukenberg.

🔗 **Link:** Not every gastric mass is epithelial. The commonest mesenchymal one is GIST.

👉 **Up next: GIST.**

2.6 Gastrointestinal Stromal Tumour, GIST (Slides 16-19)

General:

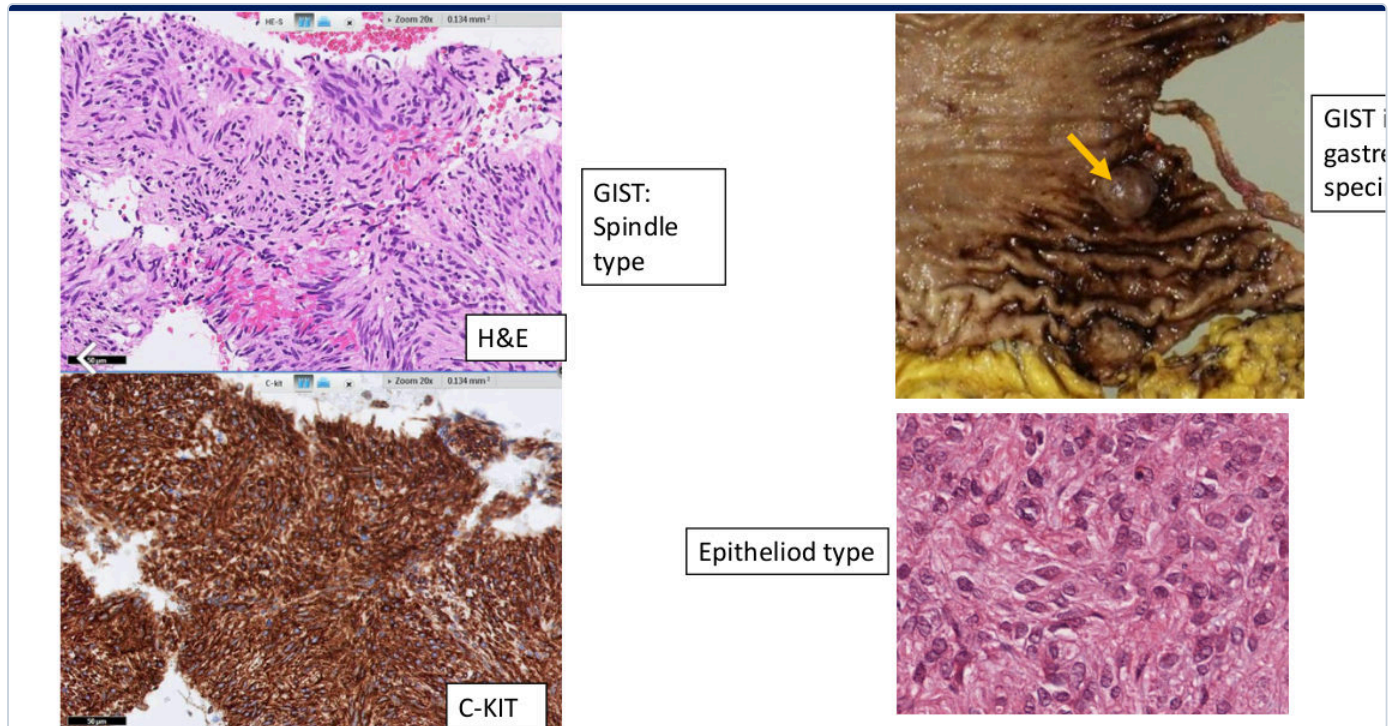
- ★ **Most common mesenchymal tumour of the abdomen**
- Anywhere along the tubular GIT; ★ **50% in the stomach**
- ★ Arises from the **interstitial cells of Cajal** (pacemaker cells)
- ★ ↑ incidence with **neurofibromatosis type 1**
- Peak age **60 y**; symptoms from **mass effect**

★★ Pathogenesis:

Mutation	Frequency	Imatinib response
★ KIT (c-KIT) , gain-of-function receptor tyrosine kinase	★ 75%	★ Often responds
★ PDGFRA	8%	Often responds
Neither	Remainder	★ Generally resistant
<i>[link: KIT is a receptor tyrosine kinase, an enzyme-linked receptor type I from Physiology Lecture 06.]</i>		

Morphology:

- Up to ★ **30 cm**; **solitary, well-circumscribed, fleshy**, overlying mucosa often **ulcerated**
- Cut surface: ★ **whorled**
- Metastases: **serosal nodules** in the peritoneal cavity or **liver**
- ★ **Spindle-cell type** (thin elongated cells) vs **epithelioid type** (plump, epithelial-looking)
- ★★ **Most useful IHC marker: KIT (CD117)**



Slide 19: GIST: spindle type (H&E), diffuse C-KIT staining, gastrectomy specimen, epithelioid type

🔁 **Recap: GIST.** Cajal-cell tumour, 50% gastric, NF1-linked, 60 y. KIT 75% / PDGFRA 8% → imatinib. Whorled, up to 30 cm, spindle or epithelioid. CD117 positive.

🔗 **Link:** The last gastric tumour arises from neuroendocrine cells.

👉 **Up next: Neuroendocrine tumours (carcinoid).**

2.7 Neuroendocrine Tumours / Carcinoid (Slides 20-25)

Terminology:

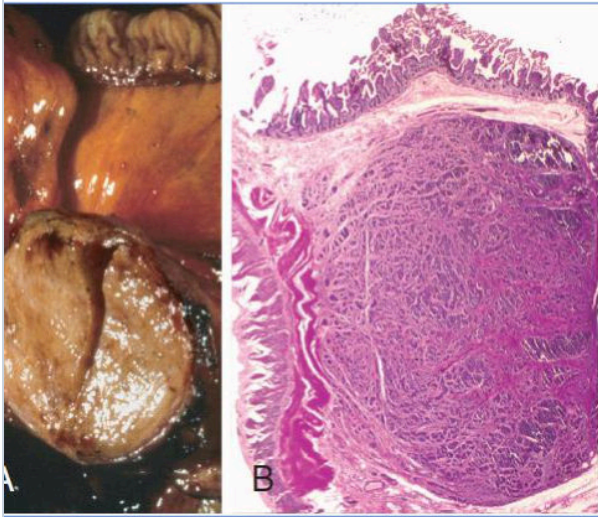
- NETs arise from **neuroendocrine cells**, found throughout the body
- ★ **Carcinoid = well-differentiated NET**, so named because it grows **more slowly than carcinomas**
- Current WHO: **low- or intermediate-grade NET**; ★ **high grade = neuroendocrine carcinoma**

Sites & associations:

- Most in the GI tract; ★ **> 40% in the small intestine**; also **tracheobronchial tree/lungs**
- ★ Gastric NETs are associated with: **endocrine cell hyperplasia**, ★ **autoimmune chronic atrophic gastritis**, ★ **MEN1**, ★ **Zollinger-Ellison syndrome**

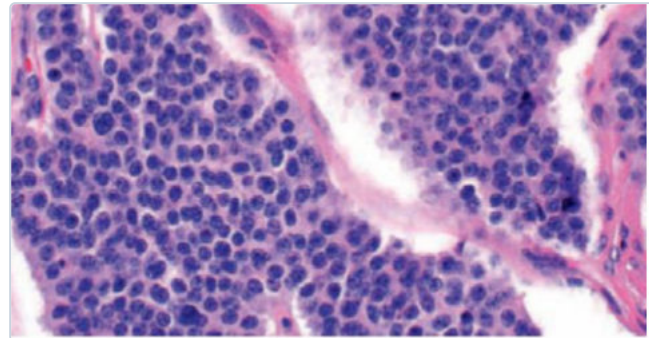
Morphology:

- Gross: **polypoid intramural or submucosal mass**; in the stomach arises within **oxyntic mucosa**; mucosa intact or ulcerated; ★ **yellow or tan**; ★ **firm** (intense **desmoplasia**) → may **obstruct** the small bowel; intestinal tumours may invade the **mesentery**
- Histology: **islands, trabeculae, strands, glands, sheets** of **uniform cells**, scant **pink granular cytoplasm**, round-oval nucleus with ★★ **"salt-and-pepper" chromatin**
- ★★ IHC: **synaptophysin** and **chromogranin A**



Gross cross section of a submucosal tumor nodule. (B) Microscopically the nodule is composed of tumor cells embedded in dense fibrous tissue.

Slide 22: Submucosal carcinoid nodule (gross) and tumour in dense fibrous tissue (micro)



Slide 23: Uniform cells with finely speckled "salt-and-pepper" chromatin

Clinical features:

- Most common in the **6th decade** (any age)
- Symptoms reflect **secreted hormones**; ★ **gastrin-producing** → **Zollinger-Ellison syndrome**
- ★★ **Carcinoid syndrome** (< 10% of patients): vasoactive substances in the **systemic circulation** → ★ **flushing**, sweating, ★ **bronchospasm**, colicky pain, ★ **diarrhoea**, ★ **right-sided cardiac valve fibrosis**, facial/chest cyanosis, intermittent hypertension, palpitations
[added: usually requires liver metastases, since the liver otherwise inactivates serotonin; the lung inactivates it before the left heart, which is why valve disease is right-sided.]

★★ Behaviour by embryological origin (Slide 25):

Origin	Site	Behaviour
Foregut	Stomach, duodenum	★ Rarely metastasise; cured by resection
Midgut	Jejunum, ileum	★ Often multiple; aggressive
Hindgut	Appendix, colorectum	★ Almost always benign

🔄 **Recap: NET.** Carcinoid = slow, well-differentiated; high grade = NE carcinoma. Gastric NETs are linked to atrophic gastritis, MEN1, ZES. Yellow firm submucosal nodules; salt-and-pepper chromatin; synaptophysin/chromogranin A. Carcinoid syndrome (< 10%): flushing, diarrhoea, bronchospasm, right-heart valves. Midgut aggressive; foregut and hindgut indolent.

2.8 Summary (Slide 26)

- **Adenocarcinoma** = most common gastric malignancy (> **90%**)
- **Intestinal vs diffuse** types with distinct morphology
- **H. pylori** = main cause of **non-cardia** cancer; also **EBV, smoking, heavy alcohol**
- **Intestinal** = gland-forming; **diffuse** = discohesive **signet-ring** cells, may cause **linitis plastica**
- **Depth of invasion + nodal/distant metastases** = most powerful prognostic indicators

- **GIST**: interstitial cells of Cajal, often gastric, frequent **KIT** mutations; **spindle or epithelioid**; **CD117** = best IHC marker
- **NETs**: **salt-and-pepper** chromatin; **synaptophysin, chromogranin A** positive

3. High-Yield Comparison Tables

Table A: Intestinal vs diffuse adenocarcinoma

	Intestinal	Diffuse
Gross	Bulky, exophytic or ulcerated mass	No mass; linitis plastica
Micro	Glands , apical mucin	Signet-ring , discohesive, no glands
Key gene	CIN (APC, SMAD4/2, TP53), MSI (MLH1), EBV	★ CDH1 / E-cadherin (hereditary: α -catenin)
Genome	Unstable, many mutations	Stable , fewer mutations
Precursor	Dysplasia, adenoma	None
Geography	High-risk areas	Uniform
Demography	Mean 55 y, M:F 2:1	[added: younger, F $\approx$ M]
Site	Antrum, lesser curve	Anywhere / whole wall

Table B: Molecular subtypes of intestinal type (see §2.3)

Table C: GIST vs NET vs adenocarcinoma

	Adenocarcinoma	GIST	NET
Cell of origin	Gastric epithelium	Interstitial cells of Cajal	Neuroendocrine cells
Key marker	[added: CK, CDX2 intestinal]	★ CD117 (KIT)	★ Synaptophysin, chromogranin A
Key histology	Glands / signet ring	Spindle / epithelioid , whorled	Salt-and-pepper chromatin
Gene	CDH1, CIN, MSI, EBV	KIT, PDGFRA	[added: MEN1 in syndromic]
Targeted drug	[added: trastuzumab (HER2+), PD-1 inhibitors]	★ Imatinib	[added: somatostatin analogues]
Associations	H. pylori, EBV, smoking, alcohol	NF1	Atrophic gastritis, MEN1, ZES

Table D: Metastatic eponyms (see §2.5)

Table E: NET behaviour by gut origin (see §2.7)

4. Rules, Exceptions, Patterns

- **> 90% of gastric cancer = adenocarcinoma.**
- **H. pylori → distal (non-cardia) cancer.**
- **Glands = intestinal; signet-ring + no glands = diffuse.**
- **CDH1 / E-cadherin loss → discohesive cells → diffuse type.**
- **Intestinal has precursors; diffuse has none.**

- **Prognosis = stage** (depth, nodes, metastases), not histological type alone.
- **Early gastric cancer can be cured (> 90%) even with positive nodes.**
- **CD117 = GIST. Synaptophysin/chromogranin = NET.**
- **KIT/PDGFR α -mutant GIST $\rightarrow$ imatinib-sensitive; wild-type $\rightarrow$ resistant.**
- **Midgut NETs are the aggressive ones.**
- **Carcinoid syndrome is uncommon (< 10%).**

5. Clinical Correlations

- **Left supraclavicular node in a patient with weight loss** $\rightarrow$ think gastric cancer (Virchow node; Troisier's sign *[added]*).
- **Bilateral ovarian masses with signet-ring cells** $\rightarrow$ Krukenberg tumour from a gastric primary.
- **Umbilical nodule** $\rightarrow$ Sister Mary Joseph nodule (peritoneal spread).
- **H. pylori eradication** reduces gastric cancer risk *[added]*.
- **Hereditary diffuse gastric cancer (CDH1 germline)** $\rightarrow$ prophylactic gastrectomy considered *[added]*; also $\uparrow$ lobular breast cancer *[added]*.
- **GIST on imatinib:** KIT/PDGFR α testing predicts response.
- **Flushing + diarrhoea + wheeze + tricuspid/pulmonary valve disease** $\rightarrow$ carcinoid syndrome (check urinary 5-HIAA *[added]*).
- **Pernicious anaemia / autoimmune atrophic gastritis** $\rightarrow$ $\uparrow$ gastrin $\rightarrow$ ECL-cell hyperplasia $\rightarrow$ gastric NET *[mechanism added]*.

6. Rapid Revision (Last-Day)

- Gastric tumours: **adenoca (> 90%)**, MALT lymphoma, leiomyosarcoma, **GIST, NET**.
- Early: dyspepsia, dysphagia, nausea. Late: weight loss, anorexia, anaemia, bleeding.
- **Japan, Chile, Costa Rica, E. Europe (20 $\times$); 55–80 y; M:F 2-3:1.**
- Causes: **H. pylori (non-cardia), EBV 10%**, smoking, alcohol; salt/preserved meat, pernicious anaemia, prior surgery, **blood group A**.
- Diffuse: **CDH1 $\rightarrow$ E-cadherin** (30% hereditary, 50% sporadic); hereditary **α -catenin**; genomically stable.
- Intestinal: **CIN** (APC, SMAD4/2, TP53; antrum) $\cdot$ **MSI** (MLH1; body) $\cdot$ **EBV** (CDKN2A, PIK3CA, PD-L1/2; fundus).
- Intestinal morphology: **antrum, lesser curve**, bulky/ulcerated, **glands**, mucin lakes.
- Diffuse morphology: **linitis plastica, signet-ring**, discohesive, **no glands**.
- Intestinal: high-risk areas, **precursors, 55 y, 2:1**. Diffuse: uniform, **no precursors**.
- Prognosis: **depth + nodes + metastases**. Early **> 90%**, advanced **< 20%** 5-yr survival.
- Mets: **Virchow, Sister Mary Joseph, Krukenberg**, left axillary, **pouch of Douglas**.
- **GIST:** most common abdominal mesenchymal tumour; **50% stomach; Cajal cells; NF1**; 60 y; **KIT 75%, PDGFR α 8%** $\rightarrow$ **imatinib**; up to 30 cm, **whorled**; spindle/epithelioid; **CD117**.
- **NET:** carcinoid = well-differentiated; high grade = NE carcinoma; **> 40% small bowel**; gastric with **atrophic gastritis, MEN1, ZES**; yellow, firm; **salt-and-pepper; synaptophysin, chromogranin A**; 6th decade.
- **Carcinoid syndrome < 10%:** flushing, sweating, bronchospasm, pain, diarrhoea, **right-sided valve fibrosis**.
- **Foregut rarely spreads $\cdot$ midgut aggressive $\cdot$ hindgut benign.**

7. Exam-Style Questions

Q1 (MCQ, lecture case, Slide 27). 53 ♀, 5 months of nausea, vomiting, epigastric pain. Barium study: **gastric outlet obstruction**. Endoscopy: **2 × 4 cm ulcerated mass at the pylorus**. Most likely biopsy finding? (A) Non-Hodgkin lymphoma (B) Neuroendocrine carcinoma (C) Squamous cell carcinoma **(D) Adenocarcinoma** (E) Leiomyosarcoma **Answer: D.** [reasoning added; the slide gives no key] Adenocarcinoma is > 90% of gastric malignancies, and the intestinal type favours the **distal stomach (antrum/pylorus)** as an **ulcerated mass**. Squamous carcinoma belongs to the oesophagus; lymphoma, NET and leiomyosarcoma are much rarer.

Q2 (MCQ). A gastrectomy shows a thick rigid wall with flattened rugae; microscopy shows single cells with mucin pushing the nucleus aside. The key molecular event is: a) KIT mutation **b) Loss of E-cadherin (CDH1)** c) MLH1 silencing d) APC mutation **Answer: b.** Linitis plastica + signet-ring cells = diffuse type.

Q3 (MCQ). A gastric spindle-cell tumour with a whorled cut surface. Best IHC marker? a) Chromogranin A b) Cytokeratin **c) CD117 (KIT)** d) Synaptophysin **Answer: c.** GIST.

Q4 (MCQ). Which feature is the most powerful prognostic indicator in gastric adenocarcinoma? a) Histological type b) Tumour size **c) Depth of invasion and nodal/distant metastases** d) Patient age **Answer: c.**

Q5 (Short answer). A patient has flushing, diarrhoea, wheeze and a tricuspid valve lesion. Name the syndrome, its histology hallmark and IHC markers, and the gut segment whose tumours are most aggressive. **Answer: Carcinoid syndrome** from a **neuroendocrine tumour**; histology: **uniform cells with "salt-and-pepper" chromatin**; IHC: **synaptophysin, chromogranin A**. **Midgut** (jejunum/ileum) NETs are the most aggressive.

8. Exam Pitfalls & Traps

- Intestinal ↔ diffuse features swapped.** Glands and precursors = intestinal; signet-ring, no glands, no precursors, CDH1 = diffuse.
- Diffuse type is genomically unstable.** No, it's **stable** with **fewer** mutations.
- H. pylori causes cardia cancer.** The lecture says **non-cardia (distal)**.
- Early gastric cancer with nodes has a poor prognosis.** No: **> 90%** 5-year survival after resection.
- GIST comes from smooth muscle.** No, from **interstitial cells of Cajal**; marker **CD117**.
- All GISTs respond to imatinib.** Only **KIT/PDGFRα-mutant** ones generally do.
- NET marker confusion:** NET = **synaptophysin/chromogranin**; GIST = **CD117**.
- Carcinoid syndrome is common.** It occurs in **< 10%**.
- Most aggressive NETs are foregut.** No: **midgut**; hindgut almost always benign.
- ⚠ M:F ratio:** Slide 4 says **2-3 : 1** overall; Slide 13 says **2 : 1** for the intestinal type. Use the one that matches the question.
- ⚠ Slide typo:** "Leiomyosarcoma" = **leiomyosarcoma**. MALT lymphoma is listed but not discussed further.